

APPLIED MACHINE LEARNING SEMESTER PROJECT REPORT

Project Title

Leukocyte Classification

Improved leukocyte classification in bone marrow cytology using convolutional neural network with contrast enhancement

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ABSTRACT

Leukocyte (white blood cell) classification plays a crucial role in the diagnosis and monitoring of various blood-related disorders, infections, and immune system diseases. Manual examination of leukocytes through microscopy is time-consuming, labor-intensive, and prone to human error, creating a need for automated and accurate classification systems. This project reproduced a transfer learning-based leukocyte classification framework that combines Contrast Limited Adaptive Histogram Equalization (CLAHE) with deep convolutional neural networks (CNNs) to improve classification performance.

Three pretrained CNN architectures, namely AlexNet, VGG16, and GoogLeNet, were implemented using the PyTorch framework and trained on a leukocyte image dataset containing three classes: LYT (Lymphocytes), MON (Monocytes), and NGS (Neutrophils). CLAHE was applied as an image enhancement technique to improve contrast and highlight important cellular features before training. Transfer learning was employed by replacing the final classification layers of the pretrained networks and fine-tuning selected layers for domain adaptation.

Several experiments were conducted to evaluate model performance and investigate the impact of fine-tuning. In particular, GoogLeNet was trained through multiple stages, including baseline transfer learning, extended training, and selective fine-tuning of the final Inception modules. The experimental results showed that AlexNet achieved the highest test accuracy of 91%, followed by VGG16 with 90%, while the fine-tuned GoogLeNet model achieved 86.22% accuracy, improving significantly from its initial performance of 73.11%. The findings demonstrate the effectiveness of transfer learning and CLAHE-based image enhancement for leukocyte classification and highlight the importance of model-specific fine-tuning strategies in improving classification accuracy under limited computational resources.

Keywords

Leukocyte Classification, Transfer Learning, Convolutional Neural Networks, CLAHE, Medical Image Analysis

1. INTRODUCTION

1.1 Problem Statement

White Blood Cells (WBCs), also known as leukocytes, are a vital part of the human immune system and play an important role in protecting the body against infections and diseases. Accurate identification and classification of different leukocyte types are essential for diagnosing various medical conditions, including leukemia, immune disorders, and blood-related diseases.

Traditionally, leukocyte classification is performed manually using microscopic examination, which is time-consuming and prone to human error. With the advancement of Artificial Intelligence and medical imaging, deep learning-based approaches have emerged as effective solutions for automating leukocyte classification. Among these approaches, Convolutional Neural Networks (CNNs) have shown promising results in image classification tasks due to their ability to automatically learn relevant image features.

1.2 Motivation

Recent studies have demonstrated that transfer learning combined with image enhancement techniques can significantly improve leukocyte classification performance. The selected research paper achieved high classification accuracy by applying Contrast Limited Adaptive Histogram Equalization (CLAHE) and transfer learning on pre-trained CNN models.

The motivation behind this project was to reproduce the proposed framework and evaluate its performance under limited computational resources using Google Colab. Additionally, the project aimed to investigate the effectiveness of transfer learning and fine-tuning techniques for improving leukocyte classification accuracy.

1.3 Objectives

The specific objectives of this project are:

- To reproduce the leukocyte classification framework proposed in the selected research paper.
- To implement and evaluate AlexNet, VGG16, and GoogLeNet using transfer learning.
- To apply CLAHE-based image enhancement and analyze its effect on classification performance.
- To investigate the impact of fine-tuning on model performance.
- To compare the obtained results with those reported in the original research paper.

1.4 Report Structure

The remainder of this report is organized as follows. Section 2 reviews related literature and existing approaches for leukocyte classification. Section 3 analyzes the selected research paper and its methodology. Section 4 describes the dataset and preprocessing techniques. Section 5 presents the implementation methodology and proposed improvements. Section 6 reports the experimental results, while Section 7 discusses the findings and limitations. Finally, Section 8 concludes the report and outlines future work.

. LITERATURE REVIEW

2.1 Traditional Approaches

Before deep learning showed up, leukocyte classification was mostly done with the usual image processing pipeline plus some classic machine learning. In practice, these approaches usually depend on hand crafted attributes, like cell outline, surface texture, color, and those morphological cues that you can pull from microscopic images. Then, for the actual decision part, methods such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), Decision Trees, and Random Forests were often chosen, to separate the different leukocyte kinds.

Even if the results looked “ok” at first, the whole thing worked because the engineered features were good. Still, feature extraction needed serious domain experience, and in many cases it simply didn’t generalize nicely across datasets, especially when the imaging conditions changed a bit. So, in the end, the traditional machine learning routes often had a hard time reaching the level of accuracy and sturdiness that clinicians really want.

2.2 Deep Learning / Modern Approaches

Then deep learning came along and, especially with Convolutional Neural Networks (CNNs), medical image classification got a lot better. In contrast to older strategies, CNNs can learn layered, nested feature descriptions straight from the pixels, which means you don’t really need that much manual feature engineering anymore.

A bunch of CNN designs have been tried successfully for image classification, such as AlexNet, VGG16, GoogLeNet, DenseNet, and InceptionResNetV2. These networks tend to perform well in medical imaging scenarios, mainly because they are able to recognize complicated visual structures and morphological signals. On top of that, transfer learning has become pretty common in medical image analysis, where pretrained CNN models are adapted for a particular specialized dataset, with less effort and usually better outcomes.

Recent studies have also explored image enhancement techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve image quality before training deep learning models. Enhanced image contrast can improve feature visibility and contribute to higher classification accuracy, particularly in medical imaging applications where subtle visual details are important.

2.3 Positioning of the Base Paper

The research paper selected for this project presents a transfer learning-based framework for leukocyte classification using multiple pretrained CNN architectures, including AlexNet, VGG16, GoogLeNet, DenseNet, and InceptionResNetV2. The study combines deep learning with CLAHE-based image enhancement to improve the quality of leukocyte images prior to classification.

A key contribution of the paper is the investigation of the impact of CLAHE on classification performance. The authors reported that applying CLAHE significantly improved overall classification accuracy, increasing performance from approximately 81% to 96.5%. These results demonstrate the effectiveness of combining image enhancement techniques with transfer learning for automated leukocyte classification.

This project builds upon the methodology proposed in the base paper by reproducing selected CNN models under limited computational resources. While the original study utilized a larger dataset and extensive training configurations, this project evaluates the practicality of achieving competitive performance using Google Colab resources and a reduced leukocyte dataset. Additionally, fine-tuning strategies are explored to further improve classification performance.

3. BASE PAPER ANALYSIS

Paper Title	<i>Leukocyte Classification Using Transfer Learning Empowered with Contrast Limited Adaptive Histogram Equalization (CLAHE)</i>
Authors	<i>Shahid Mehmood, Tariq Shahzad, Muhammad Zubair, Farman Matloob Khan, Muhammad Adnan Khan, Khmaies Ouahada & Amir H. Gandomi</i>
Venue / Year	<i>Scientific Reports, 2025</i>
DOI / URL	<i>Improved leukocyte classification in bone marrow cytology using convolutional neural network with contrast enhancement Scientific Reports</i>

3.1 Core Contribution of the Paper

The primary contribution of the selected research paper is the development of an automated leukocyte classification framework that combines transfer learning with Contrast Limited Adaptive Histogram Equalization (CLAHE). The proposed approach utilizes multiple pre-trained Convolutional Neural Network (CNN) architectures to classify different types of white blood cells while reducing training complexity through transfer learning.

A key innovation of the study is the application of CLAHE as an image enhancement technique before model training. The authors demonstrated that improving image contrast significantly enhanced feature visibility and classification performance. Experimental results showed that the proposed framework improved overall classification accuracy from approximately 81% to 96.5%, highlighting the effectiveness of combining image preprocessing and transfer learning for medical image classification.

3.2 Model Architecture

The proposed framework consists of three major stages:

1. Data Preprocessing Layer

The dataset is first prepared by splitting images into training and testing sets. To address class imbalance, the authors performed down-sampling and up-sampling operations to obtain approximately equal numbers of samples per class. Image enhancement is then applied using CLAHE to improve contrast and highlight important cellular features.

2. Training Layer

The study employs transfer learning using five pre-trained CNN architectures:

- AlexNet
- VGG16
- GoogLeNet
- DenseNet
- InceptionResNetV2

These architectures were selected due to their proven effectiveness in image classification tasks. The networks consist of common deep learning components including:

- Input Layer
- Convolutional Layers
- Activation Functions (ReLU)
- Pooling Layers
- Fully Connected Layers
- Output Layer with Softmax Classification

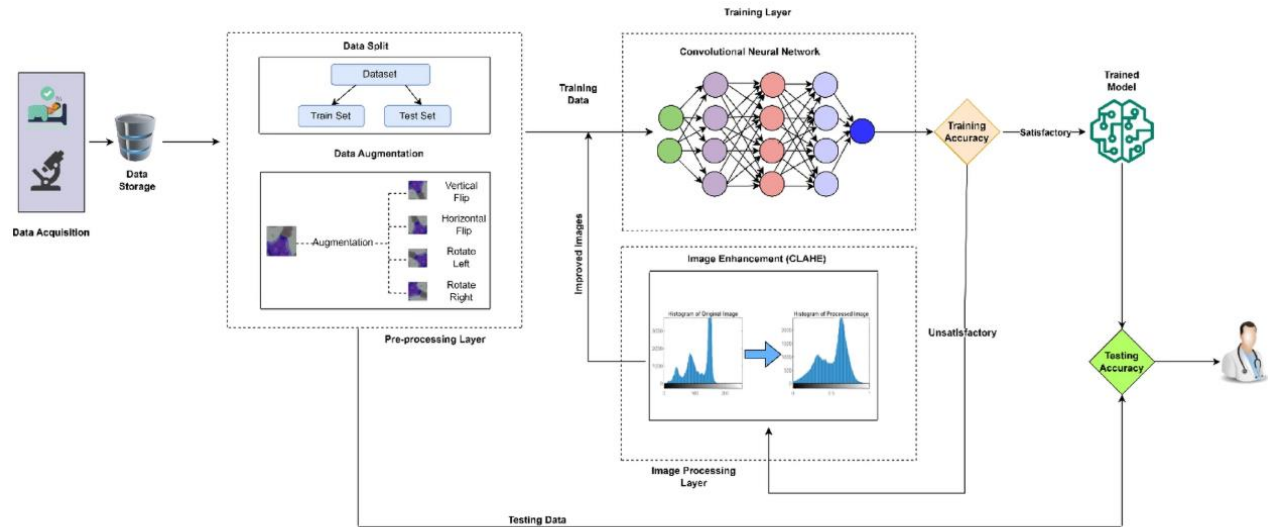
Transfer learning enables the reuse of knowledge learned from large-scale image datasets, reducing training time and computational requirements.

3. Image Processing Layer

The final stage applies Contrast Limited Adaptive Histogram Equalization (CLAHE) to improve image quality before training. CLAHE enhances local contrast while preventing excessive noise amplification, making it particularly suitable for medical image analysis where subtle visual features are important.

CLAHE Parameters Used in the Paper:

Parameter	Value
Clip Limit	2.0
Tile Grid Size	8 × 8



Proposed model for leukocyte classification.

Figure 1.1

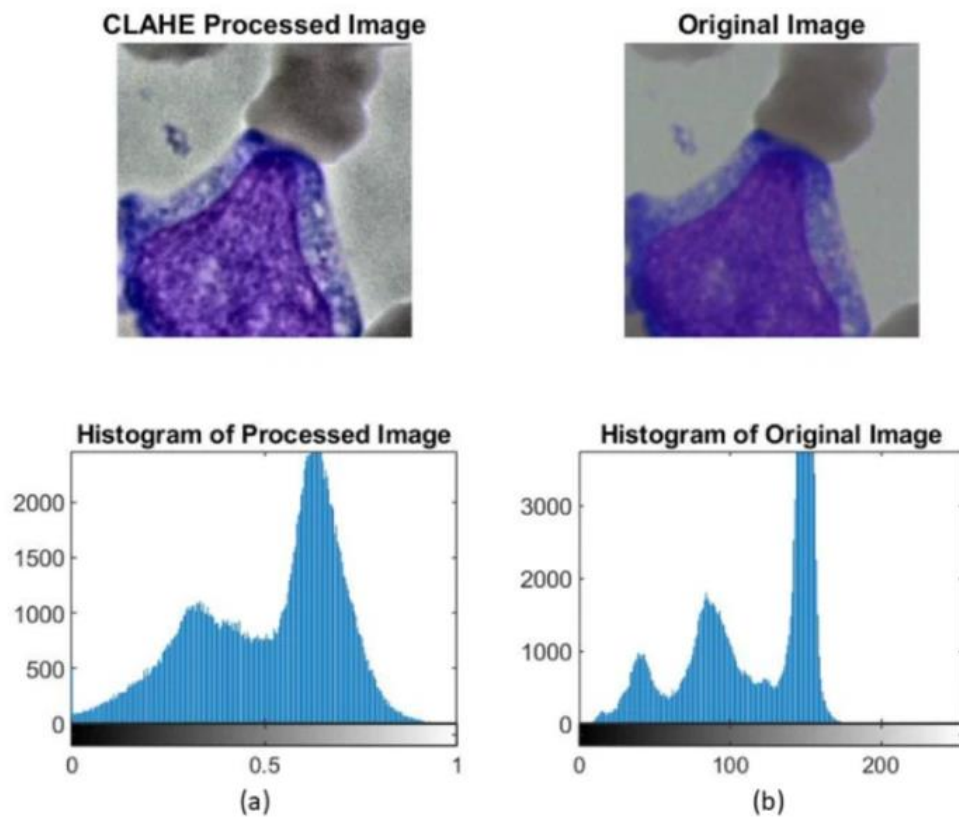


Figure 1.2

3.3 Training Procedure

Optimizer	Adam
Learning Rate	1e^-3
Batch Size	128
Epochs / Steps	40
Loss Function	Not explicitly stated in the paper (likely Cross-Entropy Loss)
Regularisation	Dropout (0.5)
Hardware (Paper)	No GPU mentioned

3.4 Reported Results

The paper evaluated five pre-trained CNN architectures before and after applying CLAHE. The results demonstrated that image enhancement significantly improved classification performance.

Results Before CLAHE

Model	Accuracy (%)
AlexNet	78.4
VGG16	81.0
GoogLeNet	77.8
DenseNet	77.4
InceptionResNetV2	77.1

Results After CLAHE

The application of CLAHE resulted in substantial performance improvements across all evaluated models. The best-performing model achieved an overall classification accuracy of **96.5%**, representing a **15.5% improvement** over the baseline performance.

The results indicate that image contrast enhancement can significantly improve feature extraction and classification accuracy in leukocyte image analysis. Consequently, the paper concludes that combining CLAHE with transfer learning provides an effective framework for automated leukocyte classification.

4. DATASET

4.1 Dataset Description

Dataset Name	Bone-Marrow-Cytomorphology_MLL_Helmholtz_Fraunhofer
Source / URL	link
Task Type	Multi-Class Image Classification
Total Samples	171,375
Training Split	Approximately 4,500
Validation Split	900
Test Split	600
Number of Classes	3
Classes Used	LYT, MON , NGS
Input Features	[e.g., 32×32 RGB images / 768-dim embeddings / ...]
Class Balance	[Balanced / Imbalanced — describe ratio]

4.2 Preprocessing Steps

To maintain consistency with the methodology proposed in the research paper, several preprocessing operations were performed before model training.

Step 1: Image Enhancement Using CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to improve image contrast and enhance cellular features. The images were converted from the BGR color space to the LAB color space, and CLAHE was applied to the luminance (L) channel before reconstructing the enhanced image.

The following parameters were used:

Parameter	Value
Clip Limit	2.0
Tile Grid Size	8 × 8

Step 2: Image Resizing

Since different CNN architectures require specific input dimensions, all images were resized before training.

Model	Input Size
AlexNet	227 × 227
VGG16	224 × 224
GoogLeNet	224 × 224

Image resizing was performed using OpenCV's interpolation techniques to maintain image quality while ensuring compatibility with the selected architectures.

Step 3: Dataset Splitting

The dataset was divided into training, validation, and testing subsets. The training set was used for model learning, the validation set was used for hyperparameter tuning and performance monitoring, and the testing set was reserved for final evaluation.

4.3 Data Augmentation (if any)

No additional data augmentation techniques were applied during this project. The primary preprocessing strategy focused on image enhancement through CLAHE and image resizing. This approach was adopted to closely follow the methodology described in the selected research paper and to ensure fair comparison of results.

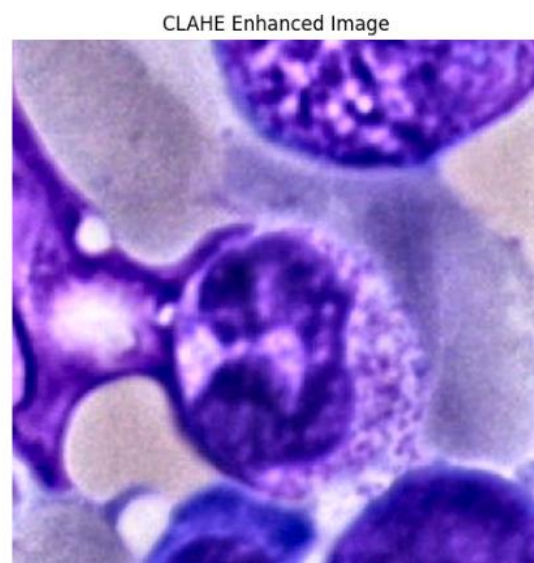
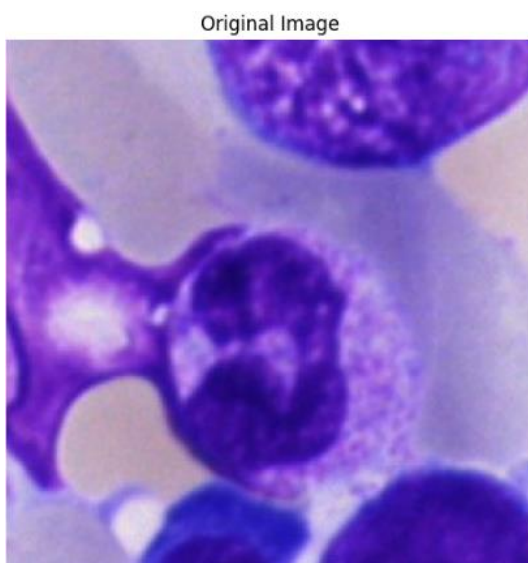
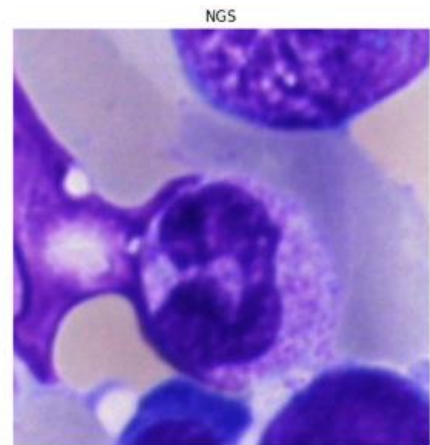
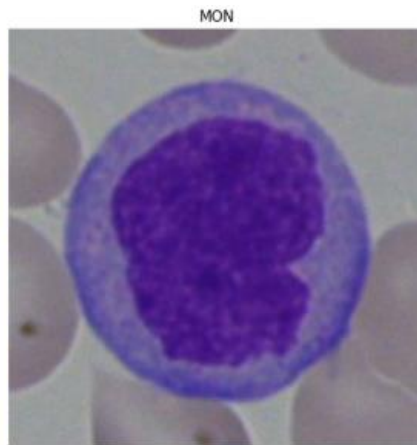
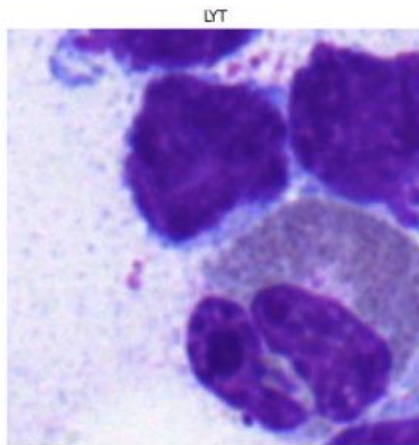
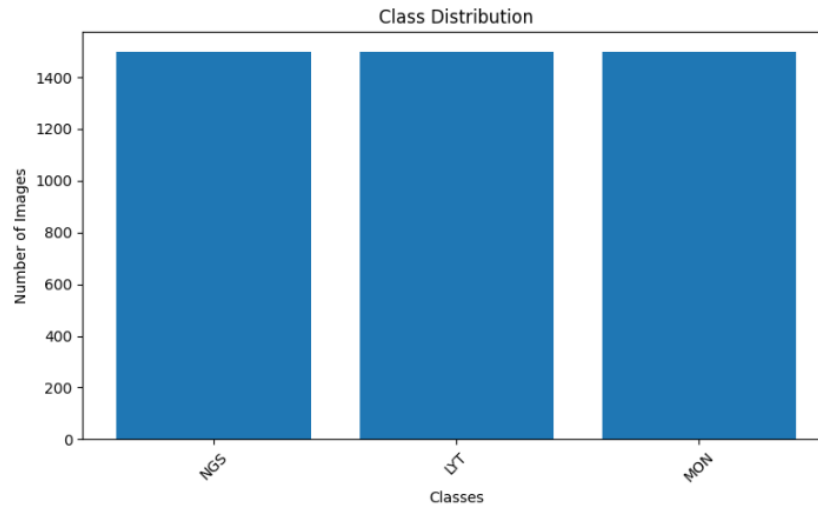
4.4 Exploratory Data Analysis (EDA)

Before model training, exploratory analysis was performed to better understand the dataset characteristics.

The following observations were made:

- The selected dataset contains three leukocyte classes: LYT, MON, and NGS.
 - The class distribution is approximately balanced, reducing the risk of classification bias toward any single class.
 - CLAHE enhancement improved image contrast and increased the visibility of cellular structures.
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- Visual inspection of sample images showed clear morphological differences among the three leukocyte classes, making them suitable for CNN-based classification.



5. METHODOLOGY

5.1 Reproduction Strategy

The objective of this project was to reproduce the leukocyte classification framework proposed in the selected research paper using transfer learning-based Convolutional Neural Networks (CNNs). The same preprocessing pipeline described in the paper was adopted, including CLAHE image enhancement and image resizing according to the requirements of each CNN architecture.

Three pretrained CNN models were selected for implementation and evaluation:

- AlexNet
- VGG16
- GoogLeNet

Transfer learning was employed to reduce training time and computational requirements. Instead of training the models from scratch, pretrained ImageNet weights were utilized and the final classification layers were modified to classify three leukocyte classes: LYT, MON, and NGS.

Due to limited computational resources and the use of Google Colab's free GPU environment, a reduced subset of the dataset was used compared to the original study. The performance of all three architectures was evaluated and compared using standard classification metrics.

5.2 Implementation Details

Framework	Pythorch
Hardware Used	NVIDIA Tesla T4 GPU
Training Time	~1.5 hours per 10 Epoch
Code Repository	Github link

5.2.1 AlexNet Implementation

A pretrained AlexNet model with ImageNet weights was utilized. Images were resized according to the model requirements and data augmentation techniques including random horizontal flipping, random rotation, and normalization were applied.

The original classification layer was replaced with a new fully connected layer consisting of three output neurons corresponding to the three leukocyte classes. Initially, the convolutional feature extraction layers were frozen and only the classifier layers were trained. Subsequently, selected deeper layers were unfrozen and fine-tuned using a smaller learning rate to improve adaptation to leukocyte images.

5.2.2 VGG16 Implementation

A pretrained VGG16 model was employed using transfer learning. The dataset was loaded from a compressed archive and split using a stratified 80/10/10 train-validation-test strategy to preserve class distributions and eliminate potential data leakage.

To improve generalization, data augmentation techniques including random horizontal flipping, vertical flipping, random rotation, Gaussian blur, and normalization were applied during training.

Initially, all convolutional layers were frozen. The final convolutional block of the network was then selectively unfrozen to allow adaptation of high-level features to leukocyte classification. The original classifier was modified by replacing the final layer with a Dropout (0.5) layer followed by a fully connected layer producing three output classes.

Training was performed using the Adam optimizer with separate learning rates for the classifier and the unfrozen convolutional layers. Weighted CrossEntropyLoss and L2 regularization were used to improve robustness and reduce overfitting.

5.2.3 GoogLeNet Implementation

A pretrained GoogLeNet model was loaded using ImageNet weights. The original fully connected classification layer was replaced with a custom classifier consisting of:

- Dropout (0.5)
- Fully Connected Layer (1024 → 3)

Initially, only the classifier layer was trained while all pretrained feature extraction layers remained frozen. This allowed the model to function as a feature extractor while learning leukocyte-specific decision boundaries.

The model was trained using the Adam optimizer and CrossEntropyLoss function, and performance was monitored through training and validation accuracy

5.3 Proposed Improvements

5.3.1 CLAHE-Based Image Enhancement

Rationale

The selected research paper demonstrated that image enhancement significantly improved leukocyte classification accuracy. Therefore, CLAHE was adopted as the primary preprocessing technique in this project.

Implementation

CLAHE was applied to the luminance channel of all images using a clip limit of 2.0 and a tile grid size of 8×8 before model training.

Expected Benefit

The enhancement improves local image contrast and highlights cellular structures, allowing CNN models to extract more informative features.

5.3.2 Fine-Tuning of GoogLeNet

Rationale

Initial experiments using GoogLeNet relied solely on transfer learning, where only the final classifier layer was trained. Although acceptable performance was achieved, further improvements became limited after extended training.

Implementation

The model was initially trained for 10 epochs and later extended to 20 epochs using transfer learning. To further improve classification performance, the final Inception modules (Inception5a and Inception5b) were unfrozen and trained together with the classifier layer using a reduced learning rate of 1×10^{-5} .

The experimental progression was:

Phase	Description
Phase 1	Transfer Learning (10 Epochs)
Phase 2	Transfer Learning (20 Epochs)
Phase 3	Fine-Tuning Inception5a + Inception5b + Classifier (30 Epochs)

Expected Benefit

Fine-tuning enables higher-level feature extraction layers to adapt to leukocyte morphology and learn domain-specific representations beyond the original ImageNet features.

6. EXPERIMENTAL RESULTS

6.1 Evaluation Metrics

Accuracy

Accuracy measures the proportion of correctly classified samples among all samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision measures the proportion of correctly predicted positive samples among all predicted positive samples.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall measures the proportion of correctly predicted positive samples among all actual positive samples.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

The F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

These metrics were used to evaluate the performance of the implemented CNN models.

6.2 Baseline Reproduction Results

The objective of the reproduction phase was to implement the transfer learning framework proposed in the selected research paper and evaluate its performance on a reduced leukocyte dataset. AlexNet, VGG16, and GoogLeNet were implemented and trained using the preprocessing pipeline described earlier. The obtained results are presented in the following sections

6.3 Comparative Results Table

Model / Variant	Accuracy (%)	Precision	Recall	F1 Score	Remarks
Baseline Paper (CLAHE + Transfer Learning)	96.50	N/A	N/A	N/A	Best result reported in original paper
AlexNet	91.00	0.91	0.91	0.91	Best performing reproduced model
VGG16	90.00	0.90	0.90	0.90	Partial fine-tuning with weighted loss
GoogLeNet (10 Epochs)	73.11	0.73	0.73	0.73	Initial transfer learning implementation
GoogLeNet (20 Epochs)	75.67	0.76	0.76	0.76	Extended training
GoogLeNet Fine-Tuned	86.22	0.86	0.86	0.86	Fine-tuning of Inception5a and Inception5b blocks

The original research paper reported a maximum classification accuracy of 96.5% after applying CLAHE-based image enhancement and transfer learning. Among the reproduced models, AlexNet achieved the highest test accuracy of 91%, followed by VGG16 with 90%. GoogLeNet showed significant improvement throughout the experimentation process, increasing from 73.11% in the initial implementation to 86.22% after fine-tuning the final Inception modules. The results demonstrate the effectiveness of transfer learning for leukocyte classification while highlighting the impact of model-specific fine-tuning strategies on performance improvement.

6.4 Training Curves

Training and validation curves were used to monitor the learning behavior of the implemented CNN models throughout the training process. These curves provide insight into model convergence, learning stability, and potential overfitting or underfitting issues.

For each model, training loss, validation loss, training accuracy, and validation accuracy were recorded during training. The resulting plots illustrate how model performance evolved over successive epochs.

AlexNet Training Curves

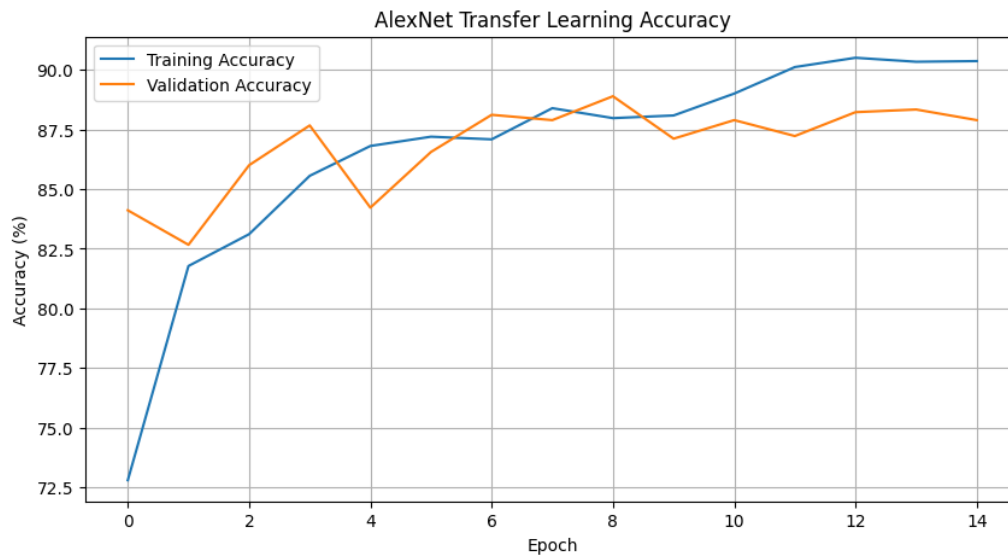


Figure 6.3: AlexNet Training and Validation Accuracy Curve

The AlexNet model demonstrated stable convergence during training, with both training and validation metrics improving consistently across epochs.

VGG16 Training Curves

Figure 6.4: VGG16 Training and Validation Loss Curve

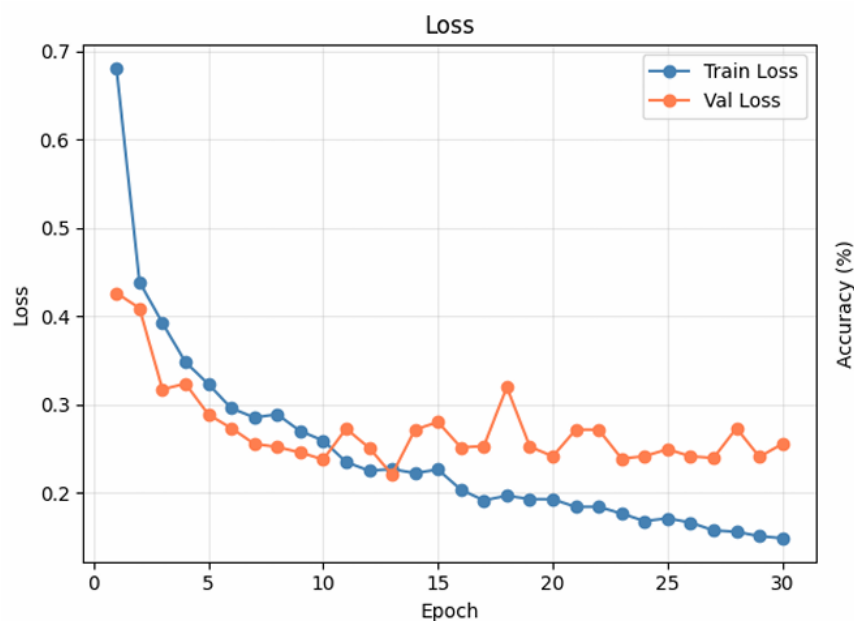
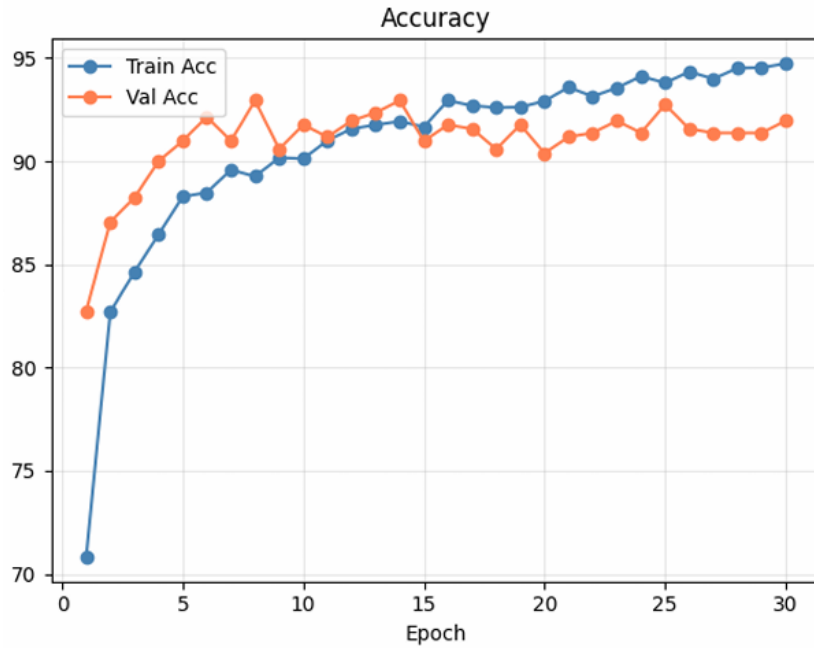
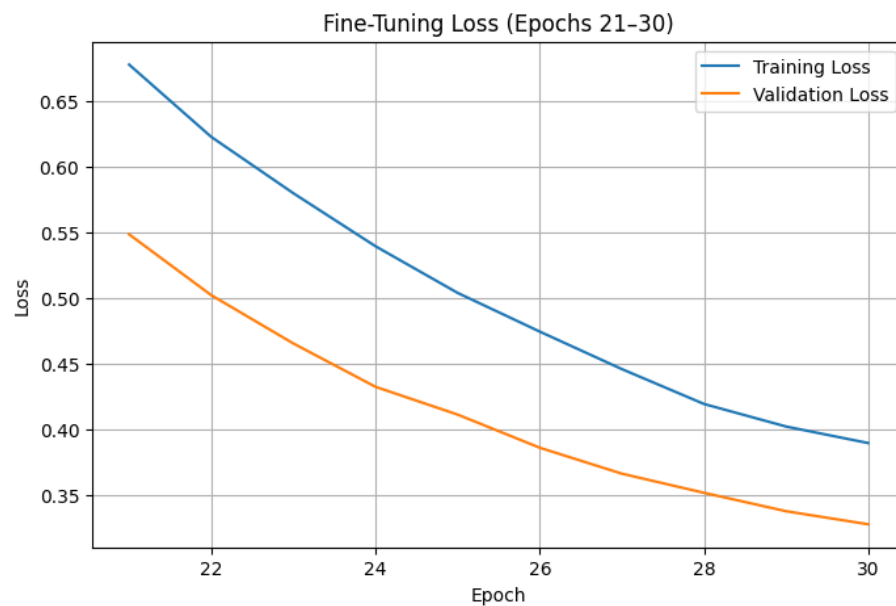
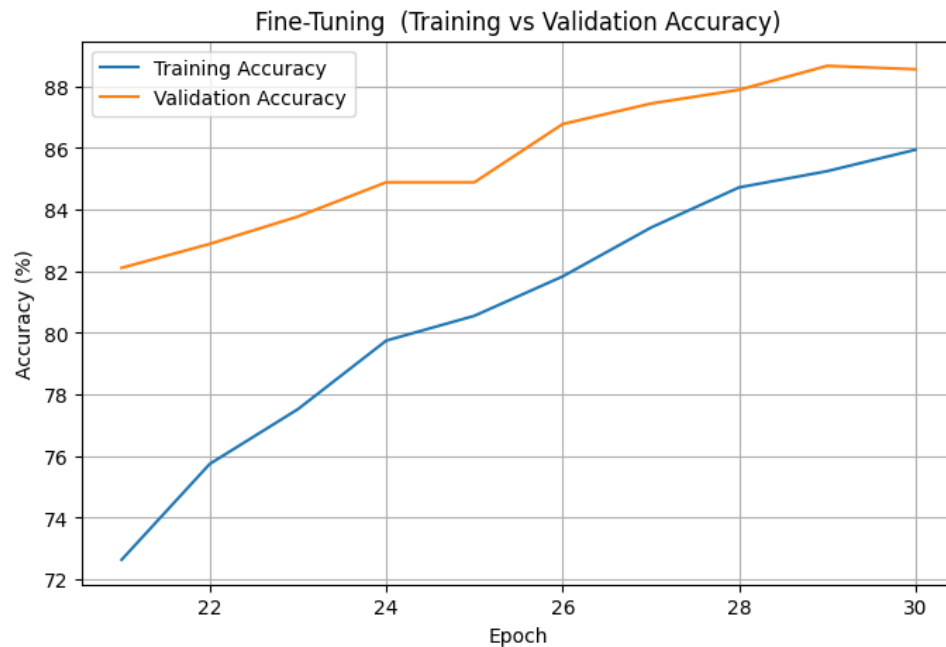


Figure 6.5: VGG16 Training and Validation Accuracy Curve

The VGG16 model showed smooth learning behavior and achieved high classification performance after fine-tuning the final convolutional block and classifier layers.

GoogLeNet Training Curves

Figure 6.6: GoogLeNet Training and Validation Loss Curve**Figure 6.7:** GoogLeNet Training and Validation Accuracy Curve



The GoogLeNet model exhibited gradual performance improvements throughout training. Significant gains were observed after fine-tuning the Inception5a and Inception5b modules, resulting in higher training and validation accuracy and reduced loss values.

6.5 Ablation Study

To evaluate the impact of the proposed fine-tuning strategy, multiple experiments were conducted using the GoogLeNet architecture. The objective was to analyze how additional training and selective layer fine-tuning affected classification performance.

Configuration	Test Accuracy (%)
GoogLeNet (10 Epochs)	73.11
GoogLeNet (20 Epochs)	75.67
GoogLeNet Fine-Tuned	86.22

The results demonstrate that increasing the number of training epochs produced a modest improvement in classification accuracy, increasing performance from 73.11% to 75.67%. However, the most significant improvement was achieved after fine-tuning the final Inception modules (Inception5a and Inception5b), which increased the test accuracy to 86.22%. This represents an overall improvement of approximately 13 percentage points compared to the initial GoogLeNet implementation, indicating that selective fine-tuning of higher-level feature extraction layers substantially improves leukocyte classification performance.

7. DISCUSSION

7.1 Interpretation of Results

The experimental results demonstrate that transfer learning is highly effective for leukocyte classification. All three implemented CNN architectures achieved strong classification performance despite being trained on a reduced dataset and limited computational resources.

Among the evaluated models, AlexNet achieved the highest test accuracy of 91%, followed closely by VGG16 with 90%. GoogLeNet initially achieved lower performance; however, after fine-tuning the final Inception modules, its test accuracy improved significantly from 73.11% to 86.22%. This improvement indicates that allowing higher-level feature extraction layers to adapt to leukocyte-specific characteristics enables the model to learn more discriminative representations and improve classification performance.

7.2 Comparison with Original Paper

The original research paper reported a maximum classification accuracy of 96.5% after applying CLAHE-based image enhancement and transfer learning. Although the reproduced models did not reach the reported performance, the obtained results are considered satisfactory given the differences in experimental conditions.

Several factors contributed to the performance gap between the original study and the reproduced models. First, the original paper used six leukocyte classes, whereas this project focused on three classes. Second, the original study was trained on a larger dataset and for 40 epochs, while the experiments conducted in this project were limited by available computational resources. Finally, the project was implemented using Google Colab's free NVIDIA Tesla T4 GPU, which restricted training time and experimentation.

Despite these limitations, the reproduced models achieved competitive performance and successfully demonstrated the effectiveness of transfer learning and CLAHE-based image enhancement for leukocyte classification.

7.3 Limitations

The following limitations were encountered during the project:

- Computational constraints limited training to Google Colab's free NVIDIA Tesla T4 GPU.
 - A reduced dataset was used compared to the original research paper.
 - Only three leukocyte classes (LYT, MON, and NGS) were included in the experiments.
 - Limited training time restricted the number of hyperparameter tuning and fine-tuning experiments that could be performed.
 - Some architectural and implementation details from the original paper were not publicly available and had to be approximated.
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7.4 Ethical Considerations

The developed models are intended to assist medical professionals in leukocyte classification and should not be used as a replacement for expert clinical diagnosis. Incorrect predictions may lead to misinterpretation of medical conditions if used without professional supervision.

Additionally, machine learning models can be affected by dataset bias and may not generalize equally well to data collected from different laboratories, imaging devices, or patient populations. Therefore, extensive validation is required before deployment in real-world clinical environments.

8. CONCLUSION & FUTURE WORK

8.1 Conclusion

This project successfully reproduced and evaluated a leukocyte classification framework based on transfer learning and CLAHE image enhancement. Three pretrained CNN architectures, namely AlexNet, VGG16, and GoogLeNet, were implemented and trained to classify three leukocyte classes: LYT, MON, and NGS.

The experimental results demonstrated that transfer learning can effectively classify leukocyte images even with limited computational resources and a reduced dataset. Among the implemented models, AlexNet achieved the highest test accuracy of 91%, followed by VGG16 with 90%. Furthermore, fine-tuning the final Inception modules of GoogLeNet significantly improved its performance from 73.11% to 86.22%, highlighting the importance of adapting pretrained feature extraction layers to domain-specific medical images.

Overall, the project successfully validated the effectiveness of CLAHE-based image enhancement and transfer learning for leukocyte classification while providing valuable insights into the performance of different CNN architectures.

8.2 Future Work

Future improvements to this work may include:

- Expanding the dataset to include all six leukocyte classes used in the original research paper.
 - Training and evaluating additional deep learning architectures such as DenseNet and InceptionResNetV2.
 - Performing extensive hyperparameter tuning to further improve classification performance.
 - Investigating advanced data augmentation techniques to improve model generalization.
 - Developing a real-time clinical decision support system for automated leukocyte classification and analysis.
-

9. REFERENCES

List all references in IEEE or APA format, numbered sequentially as they appear in the text. Minimum 10 references required.

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D. Contributions Breakdown (Group Projects)

Student Name	Contribution	% Share
Hassan	GoogLeNet implementation, fine-tuning experiments, results analysis, and report compilation	30%
Zawahir	AlexNet implementation, training, evaluation, and result analysis	17.5%
Shaheer	AlexNet implementation, training, evaluation, and result analysis	17.5%
Usman	VGG16 implementation, training, evaluation, and result analysis	17.5%

Tawab	VGG16 implementation, training, evaluation, and result analysis	17.5%
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